

Homework 3, Convex Optimization

Departamento de Matemáticas, Universidad de los Andes
Colombia

Student's name:	Luis Alejandro Rubiano
Student's code:	202013482
Email address:	la.rubiano@uniandes.edu.co
Class:	Convex Optimization
Instructor:	Mauricio Junca

1 Let K_1 and K_2 be convex cones in a vector space E . Show that $K_1 + K_2$ is a convex cone, $K_1 + K_2 \subseteq \text{co}(K_1 \cup K_2)$, if both cones contain the origin, then $K_1 + K_2 = \text{co}(K_1 \cup K_2)$. Ex. 4.3. Foundations of Optimization, Güler.

Proof: Let $k_{a1} + k_{a2}, k_{b1} + k_{b2} \in K_1 + K_2$ and $t > 0$, then $t(k_{a1} + k_{a2}) = tk_{a1} + tk_{a2}$, since $tk_{a1} \in K_1$ and $tk_{a2} \in K_2$, because K_1 and K_2 are convex cones. Therefore, $t(k_{a1} + k_{a2}) \in K_1 + K_2$.

Also $(k_{a1} + k_{a2}) + (k_{b1} + k_{b2}) = (k_{a1} + k_{b1}) + (k_{a2} + k_{b2})$, since $k_{a1} + k_{b1} \in K_1$ and $k_{a2} + k_{b2} \in K_2$, because K_1 and K_2 are convex cones. Therefore, $(k_{a1} + k_{a2}) + (k_{b1} + k_{b2}) \in K_1 + K_2$.

Thus, $K_1 + K_2$ is a convex cone.

Let $k_1 + k_2 \in K_1 + K_2$, then $k_1 + k_2 = \frac{1}{2}(2k_1) + \frac{1}{2}(2k_2)$, since $2k_1 \in K_1$ and $2k_2 \in K_2$, because K_1 and K_2 are convex cones. Therefore, since $\frac{1}{2} + \frac{1}{2} = 1$, then $k_1 + k_2 \in \text{co}(K_1 \cup K_2)$. i.e. $K_1 + K_2 \subseteq \text{co}(K_1 \cup K_2)$.

If both cones contain the origin, then $K_1 = K_1 + 0 \in K_1 + K_2$ and $K_2 = 0 + K_2 \in K_1 + K_2$. Therefore, since $K_1 + K_2$ is convex, and $\text{co}(K_1 \cup K_2)$ is the smallest convex set containing $K_1 \cup K_2$, then $K_1 + K_2 = \text{co}(K_1 \cup K_2)$. $\square$

3 Which of the following sets are convex? Ex. 2.12. Convex Optimization, Boyd.

(a) A *slab*, i.e. a set of the form $\{x \in \mathbb{R}^n \mid \alpha \leq a^T x \leq \beta\}$, where $a \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$.

Solution: Let $H_1 = \{x \in \mathbb{R}^n \mid \alpha \leq a^T x\}$ and $H_2 = \{x \in \mathbb{R}^n \mid a^T x \leq \beta\}$, then H_1 and H_2 are halfspaces, and halfspaces are convex. Therefore, the intersection of two convex sets is convex, i.e. $H_1 \cap H_2$ is convex. Thus, a slab is convex. $\square$

(b) A *rectangle*, i.e. a set of the form $\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i, i = 1, \dots, n\}$, where $\alpha_i, \beta_i \in \mathbb{R}$.

Solution: Let $H_{i1} = \{x \in \mathbb{R}^n \mid \alpha_i \leq x_i\}$ and $H_{i2} = \{x \in \mathbb{R}^n \mid x_i \leq \beta_i\}$, then H_{i1} and H_{i2} are halfspaces, and halfspaces are convex. Therefore, the intersection of two convex sets is convex, i.e. $H_{i1} \cap H_{i2}$ is convex. Thus,

$$\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i, i = 1, \dots, n\} = \bigcap_{i=1}^n (H_{i1} \cap H_{i2})$$

is convex because is the finite intersection of convex sets. $\square$

(c) A *wedge*, i.e. $\{x \in \mathbb{R}^n \mid a_1^T x \leq b_1, a_2^T x \leq b_2\}$, where $a_1, a_2 \in \mathbb{R}^n$ and $b_1, b_2 \in \mathbb{R}$.

Solution: Let $H_1 = \{x \in \mathbb{R}^n \mid a_1^T x \leq b_1\}$ and $H_2 = \{x \in \mathbb{R}^n \mid a_2^T x \leq b_2\}$, then H_1 and H_2 are halfspaces, and halfspaces are convex. Therefore, the intersection of two convex sets is convex, i.e. $H_1 \cap H_2$ is convex. Thus, a wedge is convex. $\square$

- (d) The set of points closer to a given point than a given set, i.e. $\{x \mid \|x - x_0\|_2 \leq \|x - y\|_2, \forall y \in S\}$, where $x_0 \in \mathbb{R}^n$ and $S \subseteq \mathbb{R}^n$.

Solution: For a fixed $y_0 \in S$, the set $\{x \mid \|x - x_0\|_2 \leq \|x - y_0\|_2\}$ is a halfspace, thus

$$\{x \mid \|x - x_0\|_2 \leq \|x - y\|_2, \forall y \in S\} = \bigcap_{y \in S} \{x \mid \|x - x_0\|_2 \leq \|x - y\|_2\}$$

is the intersection of halfspaces, and the intersection of convex sets is convex. Therefore, the set of points closer to a given point than a given set is convex. $\square$

- (e) The set of points closer to one set than another, i.e. $\{x \mid \text{dist}(x, S) \leq \text{dist}(x, T)\}$, where $S, T \subseteq \mathbb{R}^n$ and $\text{dist}(x, S) = \inf_{z \in S} \|x - z\|_2$.

Solution: This set is not convex in general. For example, let $S = \{0, 1\}$ and $T = \{\frac{1}{2}\}$, then the set of points closer to S than T is $\{x \mid \text{dist}(x, S) \leq \text{dist}(x, T)\} = (-\infty, \frac{1}{4}] \cup [\frac{3}{4}, \infty)$, which is not convex, because the convex in $\mathbb{R}$ are intervals. $\square$

- (f) The set $\{x \mid x + S_2 \subseteq S_1\}$, where $S_1, S_2 \subseteq \mathbb{R}^n$ with S_1 convex.

Solution: This set can be written as

$$\{x \mid x + S_2 \subseteq S_1\} = \bigcap_{y \in S_2} \{x \mid x + y \in S_1\} = \bigcap_{y \in S_2} (S_1 - y)$$

$S_1 - y$ is a translation of S_1 by $-y$, and translations of convex sets are convex. Therefore, the set $\{x \mid x + S_2 \subseteq S_1\}$ is convex because is the intersection of convex sets. $\square$

- (g) The set of points whose distance to a does not exceed a fixed fraction θ of the distance to b , i.e.

$$\{x \mid \|x - a\|_2 \leq \theta \|x - b\|_2\},$$

where $a, b \in \mathbb{R}^n$, $a \neq b$ and $0 \leq \theta \leq 1$.

Solution: Note that

$$\begin{aligned} \{x \mid \|x - a\|_2 \leq \theta \|x - b\|_2\} &= \{x \mid \|x - a\|_2^2 \leq \theta^2 \|x - b\|_2^2\} \\ &= \{x \mid x^T x - 2a^T x + a^T a \leq \theta^2 (x^T x - 2b^T x + b^T b)\} \\ &= \{x \mid x^T x - 2a^T x + a^T a \leq \theta^2 x^T x - 2\theta^2 b^T x + \theta^2 b^T b\} \\ &= \{x \mid (1 - \theta^2)x^T x - 2(a - \theta^2 b)^T x + (a^T a - \theta^2 b^T b) \leq 0\} \end{aligned}$$

When $\theta = 1$, the set is a halfspace, which is convex. When $\theta \in [0, 1)$, the set is an ball, which is convex. Therefore, the set $\{x \mid \|x - a\|_2 \leq \theta \|x - b\|_2\}$ is convex. $\square$

4 Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Show that f is non decreasing if and only if there is a convex differentiable function φ such that $f(x) = \varphi'(x)$.

Proof: $\Leftarrow$ Suppose that there is a convex function φ differentiable on $\mathbb{R}$ such that $f(x) = \varphi'(x)$.

Let $x_1, x_2, x_3, x_4 \in \mathbb{R}$ such that $x_1 \leq x_2 \leq x_3 \leq x_4$. Then, since φ is convex, we have that

$$\frac{\varphi(x_2) - \varphi(x_1)}{x_2 - x_1} \leq \frac{\varphi(x_3) - \varphi(x_2)}{x_3 - x_2} \leq \frac{\varphi(x_4) - \varphi(x_3)}{x_4 - x_3}$$

Taking the limits as $x_2 \rightarrow x_1^+$ and $x_3 \rightarrow x_4^-$, by the definition of the derivative, we have that

$$f(x_1) = \varphi'(x_1) \leq \varphi'(x_4) = f(x_4)$$

Therefore, f is non decreasing.

$\Rightarrow$ Let f be non decreasing and φ a differentiable function such that $f(x) = \varphi'(x)$, i.e. $\varphi(x) = \int_{-\infty}^x f(t) dt$. It is clear that f is integrable in the Lebesgue sense, because f is non decreasing.

Let $x_1, x_2, x_3 \in \mathbb{R}$ such that $x_1 < x_2 < x_3$. By the mean value theorem, there exists c_1, c_2 such that

$$\begin{aligned} \frac{\varphi(x_2) - \varphi(x_1)}{x_2 - x_1} &= \varphi'(c_1) = f(c_1) \\ \frac{\varphi(x_3) - \varphi(x_2)}{x_3 - x_2} &= \varphi'(c_2) = f(c_2) \end{aligned}$$

where $x_1 < c_1 < x_2 < c_2 < x_3$. Since f is non decreasing, then $f(c_1) \leq f(c_2)$, then

$$\frac{\varphi(x_2) - \varphi(x_1)}{x_2 - x_1} \leq \frac{\varphi(x_3) - \varphi(x_2)}{x_3 - x_2}$$

Therefore, φ is convex. $\square$

5 Let $f : C \rightarrow \mathbb{R}$ a twice Fréchet differentiable function on a convex set $C \subseteq \mathbb{R}^n$. The following statements are known to be equivalent:
Ex. 4.11. Foundations of Optimization, Güler.

- (a) f is convex.
- (b) $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in C$.
- (c) $\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0$ for all $x, y \in C$.
- (d) $Hf(x)$ is positive semidefinite for all $x \in C$.

Give direct proofs of

- (a) (b) $\Rightarrow$ (c)
- (b) (c) $\Rightarrow$ (b)
- (c) (c) $\Rightarrow$ (d)
- (d) (d) $\Rightarrow$ (c)

Proof:

- (a) (b) $\Rightarrow$ (c):

Starting from the condition (b),

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

Subtracting $f(x)$ from both sides, we have

$$f(y) - f(x) \geq \langle \nabla f(x), y - x \rangle$$

Now since this holds for all $x, y \in C$, it also follows that

$$f(x) - f(y) \geq \langle \nabla f(y), x - y \rangle$$

Adding the last two inequalities, we have

$$f(y) - f(x) + f(x) - f(y) \geq \langle \nabla f(x), y - x \rangle + \langle \nabla f(y), x - y \rangle$$

Therefore,

$$0 \geq \langle \nabla f(x), y - x \rangle + \langle \nabla f(y), x - y \rangle = \langle \nabla f(x) - \nabla f(y), y - x \rangle$$

Which implies

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0 \quad \square$$

(b) (c) $\Rightarrow$ (b):

Starting from the condition (c),

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0$$

Let $x_\tau := x + \tau(y - x)$, then

$$\begin{aligned} f(y) &= f(x) + \int_0^1 \langle \nabla f(x_\tau), y - x \rangle d\tau \\ &= f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \langle \nabla f(x_\tau) - \nabla f(x), y - x \rangle d\tau \\ &= f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \frac{1}{\tau} \langle \nabla f(x_\tau) - \nabla f(x), x_\tau - x \rangle d\tau \\ &\geq f(x) + \langle \nabla f(x), y - x \rangle \quad \square \end{aligned}$$

(c) (c) $\Rightarrow$ (d):

Starting from the condition (c),

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0$$

Let $y := x + th$, then

$$\langle \nabla f(x + th) - \nabla f(x), th \rangle \geq 0$$

Also, for the Hessian matrix, we have that

$$\nabla f(x + th) = \nabla f(x) + tHf(x)h + o(|t|)$$

or equivalently

$$\nabla f(x + th) - \nabla f(x) = tHf(x)h + o(|t|)$$

Therefore,

$$\langle \nabla f(x + th) - \nabla f(x), th \rangle = \langle tHf(x)h + o(|t|), th \rangle = t^2 h^T Hf(x)h + o(t^2)$$

Thus, if $Hf(x) < 0$, then the last expression is negative for a small enough t , a contradiction. Therefore, $Hf(x)$ is positive semidefinite. $\square$

(d) (d) $\Rightarrow$ (c):

Note that

$$f(y) = f(x) + \nabla f(x)^T (y - x) + \frac{1}{2} (y - x)^T Hf(x) (y - x) + o(\|y - x\|^2)$$

Since $Hf(x)$ is positive semidefinite, then

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

which is the condition (b), which was proved to be equivalent to (c). $\square$

**7 Consider the function $f(x) = \langle c, x \rangle - \sum_{j=1}^m \log(1 - \langle a_j, x \rangle) - \sum_{i=1}^n \log(1 - x_i^2)$.
Show that f is convex.**

Proof: Let $g(x) = \langle c, x \rangle$, $h(x) = \sum_{j=1}^m \log(1 - \langle a_j, x \rangle)$ and $l(x) = \sum_{i=1}^n \log(1 - x_i^2)$.

$g(x)$ is linear, and therefore convex.

The function $1 - \langle a_j, x \rangle$ is affine, and therefore concave. Since the logarithm function is concave, then $h(x)$ is concave as it is the sum of concave functions.

As a function of one variable, $\log(1 - x_i^2)$ has second derivative $-\frac{2(x_i^2 + 1)}{(1 - x_i^2)^2}$, since for all $x_i \in \mathbb{R}$ both the numerator and the denominator are positive, then $(\log(1 - x_i^2))'' < 0$, i.e. $\log(1 - x_i^2)$ is concave. Therefore, $l(x)$ is concave as it is the sum of concave functions.

Since $g(x)$ is convex, $h(x)$ is concave, thus $-h(x)$ is convex, $l(x)$ is concave, then $-l(x)$ is convex, and the sum of convex functions is convex, then $f(x) = g(x) - h(x) - l(x)$ is convex. $\square$